

Article

Differential Urban-Rural Inequalities and Driving Mechanisms of PM_{2.5} Exposure in the Central Plains Urban Agglomeration, China

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Abstract

Exposure to PM_{2.5} poses severe risks to public health and sustainable development, with exposure inequalities exacerbated by variations in atmospheric activity and uneven regional development. However, the urban-rural inequalities and natural-human driving mechanisms underlying PM_{2.5} exposure inequalities within urban agglomerations are poorly understood. Taking the Central Plains Urban Agglomeration (CPUA) in China as an example, this study investigated the spatio-temporal variations of PM_{2.5} and considered its future trends. The Theil index was employed to quantify PM_{2.5} exposure inequalities. An interpretable machine learning model (RF-SHAP) was applied to identify the raster natural and socioeconomic driving factors. We found that 99.68% of the CPUA exhibited a decreasing trend in ground-level PM_{2.5}. The overall Theil index decreased from 0.168 to 0.142, with a rural decline from 0.115 to 0.084, suggesting an overall reduction in air pollution inequalities, particularly in rural areas. Conversely, the urban Theil index increased from 0.096 to 0.208, highlighting an increasing inequality in urban PM_{2.5} exposure. Resource-based cities, such as Changzhi, Jincheng, and Jiaozuo, exhibited the largest PM_{2.5} exposure inequality. Elevation was identified as the dominant factor influencing overall and rural PM_{2.5} exposure inequalities, while population density was the primary driver of urban PM_{2.5} exposure inequalities. This study highlighted the differences in urban–rural PM_{2.5} inequalities and their drivers at the city agglomeration scale. The aims were to mitigate PM_{2.5} exposure inequalities through socio-environmental systems, provide evidence for the integrated management of PM_{2.5} exposure inequalities in city agglomerations, and support regional sustainable development.

Keywords: PM_{2.5} exposure; inequality; spatial and temporal analysis; urbanization



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1. Introduction

Atmospheric particulate matter with an aerodynamic diameter of less than or equal to $2.5\text{ }\mu\text{m}$ ($\text{PM}_{2.5}$) is the primary contributor to air pollution events, such as haze [1]. Prolonged exposure to $\text{PM}_{2.5}$ significantly increases the risk of cardiovascular, cerebrovascular, and respiratory diseases, thereby diminishing life expectancy in affected populations [2]. Various types of particles, including vehicle exhaust emissions and fugitive dust, exacerbate the $\text{PM}_{2.5}$ exposure inequalities between urban and rural areas [3,4]. This imbalance has hampered progress toward achieving the Sustainable Development Goals (SDGs) [5–9]. With the increasing pace of urbanization, the sources of $\text{PM}_{2.5}$ have become increasingly diverse, regional, and complicated, and $\text{PM}_{2.5}$ concentrations are influenced by climate conditions, geographic characteristics, pollution control technology and management, and socio-economic factors. Identifying emission sources and reducing $\text{PM}_{2.5}$ emissions, as well as implementing regional joint prevention measures, are critical strategies for improving air quality and promoting regional equity. The success of these measures in safeguarding public health and human well-being requires a comprehensive understanding of the urban–rural $\text{PM}_{2.5}$ exposure inequalities and their underlying drivers [7,10].

Several studies have explored the factors driving air pollution across varying scales. At the urban scale, meteorological datasets have been widely applied to assess the climate influences on air pollution over daily, monthly, and annual periods [11,12]. Atmospheric chemical transport models have been used during the critical pollution seasons to trace particulate matter sources and facilitate the precise attribution of pollutant exposure within localized regions [13]. On broader regional scales, studies have examined the socio-economic drivers of particulate pollution, explored the ‘urban pollution island’ effect, and developed strategies for their integrated management [14–16]. Rapid economic growth, coupled with inadequate pollution control measures, amplifies air pollution exposure, endangering public health and ecological stability [5,8]. Therefore, identifying effective pathways for mitigating and controlling air pollution, based on a comprehensive understanding of its drivers, remains a research priority.

The inequalities of air pollution exposure among different regions have received increasing attention from researchers. Addressing these inequalities, particularly amidst the ongoing economic expansion in China, is vital for improving human well-being [17–19]. Advanced geospatial analyses have quantitatively assessed exposure inequalities from the street to global scales, uncovering how inequalities in pollution exposure correlate with negative health outcomes, including mortality [7,20–24]. Additionally, studies have linked these inequalities to socio-environmental determinants such as economic development and climate change, highlighting the complex interplay between environmental exposure, health inequities, and regional development. These studies have developed strategies for advancing environmental equity and managing air pollution across regions [25–27]. However, the differences in $\text{PM}_{2.5}$ exposure inequality between urban and rural areas over a long period of time still need to be further clarified to reduce the inequality in development urban agglomeration between urban and rural areas.

Several critical research gaps have been identified. First, there is a need to explore the spatio-temporal patterns and future trends of air pollution in urban agglomerations to identify governance hotspots. Second, the long-term evolution of urban–rural $\text{PM}_{2.5}$ exposure inequality in developing countries remains poorly understood, especially in the context of policy constraints and rapid urbanization. While previous studies have examined $\text{PM}_{2.5}$ exposure inequalities, few have integrated natural, climate, emission-related, and socio-economic factors to analyze urban–rural inequalities comprehensively [28].

This study investigated urban–rural $\text{PM}_{2.5}$ exposure inequalities within China, with an emphasis on the Central Plains Urban Agglomeration (CPUA). The CPUA is a densely

populated region undergoing rapid urbanization and industrialization, thus facing significant air pollution challenges. The CPUA includes the Taihang Mountains, which influence the transport of air pollutants, leading to high levels of air pollution in this area. Furthermore, the region is characterized by a dense population and the presence of heavy industries, particularly in the northern CPUA, where the abundant coal resources exacerbate vulnerabilities in human well-being [29]. Given its distinctive natural environment and relatively slow pace of socio-economic development, the CPUA was selected as a case study to precisely identify the socio-economic and environmental drivers behind urban–rural PM_{2.5} exposure inequalities under complex terrain and climate conditions. The findings were used to develop a model for managing PM_{2.5} exposure inequalities in densely populated, industrialized regions and to promote integrated management strategies at the urban agglomeration scale. By investigating the spatio-temporal patterns of PM_{2.5} pollution and urban–rural exposure inequalities within the CPUA, the study aimed to identify priority governance zones for air pollution management and proposed scientifically grounded pathways for their integrated management. The research directly aligned with SDG 3 (Good health and well-being), SDG 10 (Reduced inequalities), and SDG 11 (Sustainable cities and communities) [9,30–32]. The study addressed three core research questions: (1) How have the spatial and temporal patterns of PM_{2.5} pollution evolved within representative urban agglomerations, and what insights can be derived from forecasting future trends and identifying pollution hotspots? (2) What are the dynamics of urban–rural PM_{2.5} exposure inequalities within urban agglomerations, and how have they changed over time? (3) How do natural, climate, and socio-economic drivers collectively influence urban–rural exposure inequalities, and what strategies can be proposed for effective mitigation?

2. Research Methods

2.1. Research Area

Figure 1 shows the location of the CPUA in China, encompassing 30 prefecture-level cities and extends across five provincial-level administrative divisions: Henan, Shandong, Anhui, Shanxi, and Hebei (Figure 1a,b). The CPUA exemplifies a typical case of rapid urbanization and socio-economic transformation, covering approximately 287,000 km²—accounting for 2.99% of China. The CPUA has a complex topography, which is dominated by mountains in the west and plains in the east-central part, with an average elevation of about 288 m (Figure 1c). The complex topography has led to differences in regional atmospheric activity and socio-economic development. In 2022, the China Urban Statistical Yearbook showed that the CPUA contributed 6.88% of the national GDP (8485 billion yuan) and accommodated 13.32% of the total population (188.03 million) of China. The annual mean PM_{2.5} concentration in the CPUA is 44 µg/m³, which is 1.46 times the average PM_{2.5} concentration in China (30.8 µg/m³). The ratio of maximum to minimum of PM_{2.5} concentrations in this area is approximately 2 (Figure 1d). These data highlight a stark disparity between urban and rural areas within the CPUA regarding air pollution exposure, socio-economic conditions, and public resource distribution. As a key component of the national “Rise of Central China” strategy, the CPUA faces substantial pressure to balance economic growth and environmental protection. The region urgently requires scientific evidence to guide pollution mitigation and promote sustainable development. Studying the CPUA can offer practical insights into improving local air pollution control, while also providing a valuable reference for pollution prevention and management in similar urban agglomerations worldwide [33] (Figure 1e,f).

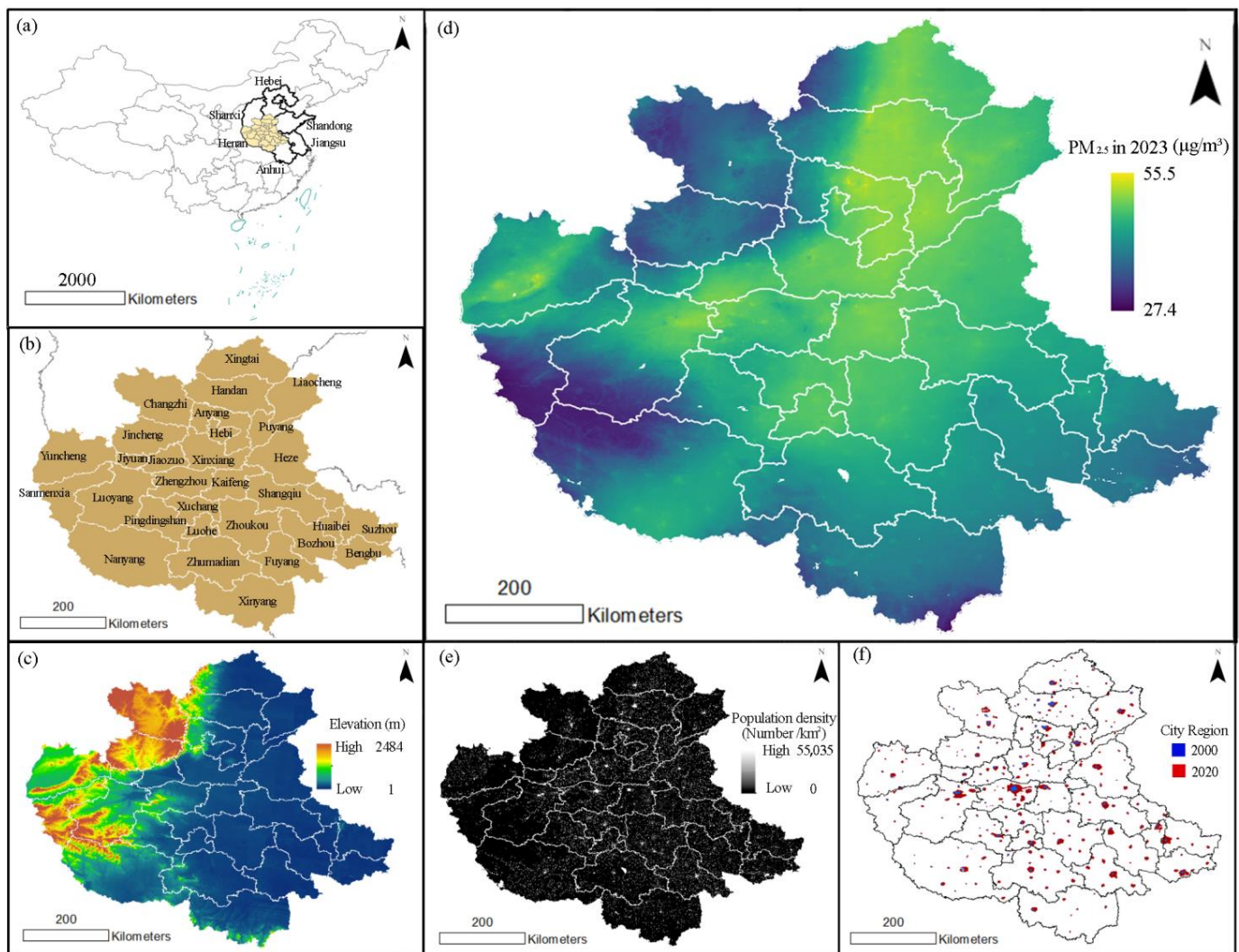


Figure 1. Map of the study area. (a). Geographic location of the Central Plains Urban Agglomeration (CPUA) in China. (b). Spatial distribution of the 30 prefecture-level cities within the CPUA. (c). Elevation of the CPUA, derived from a digital elevation model (DEM). (d). The PM_{2.5} concentration across the CPUA in 2023, mapped at a resolution of 1 × 1 km. (e). A population density map of the CPUA in 2020. (f). Urban expansion in the CPUA between 2000 and 2020, with blue representing areas built prior to 2000 and red indicating newly developed areas after 2000.

2.2. Data Preparation

2.2.1. PM_{2.5} Data

The PM_{2.5} data used in this study were sourced from the Big Data Seamless 1 km Ground-level PM_{2.5} dataset within the China High Air Pollutants (CHAP) dataset, which provides annual ground-level PM_{2.5} estimates from 2000 to 2023 at a spatial resolution of 1 km (<https://weijing-rs.github.io/> (accessed on 25 August 2025)). This dataset integrates multi-source remote sensing data and employs a four-dimensional spatiotemporal-extreme stochastic model, combining ground-based observational data, atmospheric reanalysis data, emission inventories, and model simulations. It effectively addresses the data gaps in the Moderate Resolution Imaging Spectroradiometer (MODIS) multi-angle implementation of atmospheric correction (MAIAC) aerosol optical depth (AOD) products and exhibits a high cross-validation coefficient of determination between training and test sets ($C-V R^2 = 0.92$), making it suitable for the long-term analysis of PM_{2.5} pollution trends at the urban agglomeration scale (Table 1) [34,35].

Table 1. Indicators, source, and classification of all the data used in this study (All data accessed on 25 August 2025).

Dimension	Indicator	Spatial Resolution	Source	Reference
Pollution	PM _{2.5} (PM)	1000 m	https://zenodo.org/records/13340222	[34,35]
	Urban Built Area	1000 m	https://doi.org/10.6084/m9.figshare.16602224.v1	[36]
Natural	Normalized Difference Vegetation Index (NDVI)	1000 m	https://www.earthdata.nasa.gov/	[37]
	Elevation (ELE)	1000 m	https://search.earthdata.nasa.gov/	[38]
Climate	Precipitation (PRE)	1000 m	https://data.tpdc.ac.cn/zh-hans/data	[39]
	Temperature (TEM)	1000 m	https://data.tpdc.ac.cn/zh-hans/data	[40]
Discharge	Electricity consumption (EC)	1000 m	https://doi.org/10.6084/m9.figshare.17004523.v1	[41]
	Ozone Concentration (O ₃)	1000 m	http://geodata.nnu.edu.cn/data	[42]
	Nighttime light (NTL)	1000 m	http://geodata.nnu.edu.cn/	[43]
Social-Economic	Population density (POP)	1000 m	https://www.eastview.com/resources/e-collections/landscan/	[44]
	Gross Domestic Product (GDP)	1000 m	https://www.resdc.cn/doi/doi.aspx?DOIid%20=%2033	[45]

2.2.2. Urban–Rural Boundary Data

To delineate urban and rural areas and examine the inequalities in air pollution exposure within the CPUA, the study employed an annual global urban dataset based on coordinated nighttime lighting data [36]. This dataset provides long-term global urban mapping with a consistent spatial and temporal resolution (1992–2020, 1 km) matching that of the PM_{2.5} dataset from 2000 to 2020 based on the origin remote sensing data of DMSP-OLS and NPP-VIIRS. Due to the limited temporal resolution of the urban boundary data source, the study only analyzed the inequality of urban–rural pollution exposure in the CPUA from 2000 to 2020. It dynamically captures areas of high-intensity human activity, enabling the precise identification of urban and rural areas within the CPUA and facilitating an in-depth analysis of urban–rural PM_{2.5} exposure inequality (Table 1) [36]. To ensure the accuracy of city boundaries, urban areas were precisely delineated on an annual basis. The definition of urban areas was based on data sources, namely areas of human activity with continuous nighttime light intensity.

2.2.3. Control Factor Data

To explore the variability in the drivers of urban–rural PM_{2.5} exposure inequality, climate, emission, and socio-economic factors were investigated for five representative years: 2000, 2005, 2010, 2015, and 2020. The natural factors were vegetation cover (normalized difference vegetation index, NDVI) [37] and elevation (ELE) [38]. The climate factors were precipitation (PRE) [39] and temperature (TEM) [40]. The emission factors were ground-level PM_{2.5} (PM) [34,35], electricity consumption (EC) [41], and ozone concentration (O₃) [42]. The socio-economic factors were nighttime lighting data (NTL) [43], population density (POP) data in a 1 × 1 km raster [44], and gross domestic production (GDP) data [45]. All control factors data have a space resolution of 1 km from 2000 to 2020. These variables were used to analyze urban–rural PM_{2.5} exposure inequalities and provide a foundation for integrated urban–rural air pollution management (Table 1).

2.3. Methods

2.3.1. Spatial and Temporal Trends of PM_{2.5} in the Central Plains Urban Agglomeration

The Sen trend analysis and Mann-Kendall (MK) test are widely applied techniques for the trend analysis of an atmospheric pollution concentration time series because they can overcome the interference from outliers or large measurement errors [46], providing more robust results than traditional methods [47]. The trend of PM_{2.5} exposure intensity

and its statistical significance in the CPUA from 2000 to 2023 were analyzed using these two methods. The formula for calculating the Sen slope (β) is as follows [48–50]:

$$\beta = \text{median} \frac{PM_j - PM_i}{j - i} \quad (1)$$

where “median” refers to the median of the $PM_{2.5}$ data series, with $1 < I < j < n$; i, j represents different times; and PM_i and PM_j are the corresponding $PM_{2.5}$ values at times i and j , respectively.

When the slope β is >0 , it indicates an upward trend; when $\beta = 0$, there is no trend; and when β is <0 , it means a downward trend.

The Mann-Kendall test was used to identify the year with a significant $PM_{2.5}$ change. The test is based on the statistic S , which is defined as

$$S = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \text{sgn}(x_j - x_i) \quad (2)$$

$$\text{sgn}(x_j - x_i) = \begin{cases} +1 & \text{if } (x_j - x_i) > 0 \\ 0 & \text{if } (x_j - x_i) = 0 \\ -1 & \text{if } (x_j - x_i) < 0 \end{cases} \quad (3)$$

A trend test was performed using the statistic Z . Z is calculated as follows:

$$Z = \begin{cases} \frac{S}{\sqrt{\text{Var}(S)}} & (S > 0) \\ 0 & (S = 0) \\ \frac{S+1}{\sqrt{\text{Var}(S)}} & (S < 0) \end{cases} \quad (4)$$

where Var is calculated as

$$\text{Var}(S) = \frac{n(n-1)(2n+5)}{18} \quad (5)$$

where n is the total number of data points in the series and t_i denotes the number of data points in group i . A linkage group is a set of data with the same value.

A bilateral trend test is used to assess trends at a specific α significance level. When $|Z| \leq Z_{1-\frac{\alpha}{2}}$, the null hypothesis is accepted, indicating that the trend is not significant. When $|Z| > Z_{1-\frac{\alpha}{2}}$, the null hypothesis is rejected, and the trend is considered significant. Three significance levels were tested: $\alpha = 0.01$, $\alpha = 0.05$, and $\alpha = 0.1$. If the absolute value of Z exceeded 1.65, 1.96, or 2.58, the trend was considered significant at the 90%, 95%, and 99% confidence levels, respectively. In this study, the $PM_{2.5}$ changes were categorized into eight levels (Table 2).

2.3.2. Predicted Changes in $PM_{2.5}$ in the Central Plains Urban Agglomeration

The Hurst index (H) was employed to characterize the future trends in the $PM_{2.5}$ concentration. The Hurst index is a key statistical metric that measures the persistence or self-similarity of a time series. The Hurst index is calculated as follows [51]:

First, the spatial and temporal variation sequence of $PM_{2.5}$ is defined as $PM_{(t)}$, where $t = 1, 2, 3, \dots$, and the mean series is defined as $\overline{PM}_{(\tau)}$.

$$\overline{PM}_{(\tau)} = \frac{1}{\tau} \sum_{t=1}^{\tau} PM_{(t)} \quad (6)$$

The cumulative deviation sequence $U_{(t,\tau)}$ is denoted as

$$U_{(t,\tau)} = \sum_{t=1}^{\tau} (PM_{(t)} - \overline{PM}_{(\tau)}) \quad (1 \leq t \leq \tau) \quad (7)$$

Table 2. Eight levels defining the temporal variation of PM_{2.5} intensity.

β	Z	Trend Type	Trend Features
>0	$ Z > 2.58$	4	Extremely significant increase (ESI)
	$1.96 < Z \leq 2.58$	3	Significant increase (SI)
	$1.65 < Z \leq 1.96$	2	Slightly significant increase (SSI)
	$ Z \leq 1.65$	1	No significant increase (NSI)
0	$Z = 0$	0	No change (NC)
<0	$ Z \leq 1.65$	−1	No significant reduction (NSR)
	$1.65 < Z \leq 1.96$	−2	Slightly significant reduction (SSR)
	$1.96 < Z \leq 2.58$	−3	Significant reduction (SR)
	$ Z > 2.58$	−4	Extremely significant reduction (ESR)

The extreme deviation series (R) is

$$R_{(\tau)} = \max_{1 \leq t \leq \tau} U_{(t,\tau)} - \min_{1 \leq t \leq \tau} U_{(t,\tau)} \tau = 1, 2, 3 \dots \quad (8)$$

The standard deviation series (S) is

$$S_{(\tau)} = \left[\frac{1}{\tau} \sum_{t=1}^{\tau} \left(PM_{(t)} - \overline{PM}_{(\tau)} \right)^2 \right]^{\frac{1}{2}} \quad (9)$$

Finally, the Hurst index is calculated as follows:

$$\frac{R}{S} = \frac{R_{(\tau)}}{S_{(\tau)}} = (\alpha \tau)^H \quad (10)$$

where H represents the Hurst index, ranging from 0 to 1. When $H = 0.5$, it indicates no discernible trend, denoting a state of “no significant change”; when $H > 0.5$, it suggests that the process exhibits persistence, and the future trend is likely to align with the past trend, which is known as “positive persistence”; and a value of $H < 0.5$ indicates that the future trend will likely be the opposite of the past trend, which referred to as “anti-persistence”.

In this study, the MK test and Sen’s slope estimator were combined with the Hurst index to analyze and predict the future trend of PM_{2.5} concentration changes in the CPUA.

2.3.3. Inequality of PM_{2.5} Exposure in the Central Plains Urban Agglomeration

The Theil index is widely used to assess inequalities between regions or groups, based on information theory. It has two commonly used forms: the Theil T and Theil L indices [52]. In this study, the Theil L index, which incorporates population weights, was used to calculate air pollution inequality [14,53]:

$$L = \sum_i \frac{P_i}{P} \log \left(\frac{P_i}{P} * \frac{C}{C_i} \right) \quad (11)$$

where P_i is the population corresponding to the raster; P is the total population of the region; C_i is the PM_{2.5} concentration corresponding to the raster; and C represents the total PM_{2.5} concentration within the region.

Additionally, to further quantify the contribution of intra-regional and extra-regional inequality to overall inequality, the Theil index was decomposed into within-group variation (TWG) and between-group variation (TBG) [22]. Specifically, TWG represents the inequality that exists within individual regions, reflecting disparities among counties in cities within the same broader administrative area. In contrast, TBG captures the inequality that arises between different regions, indicating the extent to which inequality of PM_{2.5} ex-

posure differs across broader regional units. This decomposition allows for a more nuanced understanding of the spatial structure of exposure inequality and helps identify whether local or broader disparities are the dominant drivers of overall inequality. To calculate TWG and TBG, PM_{2.5} data was first aggregated to the county scale. The contributions of urban inequality, rural inequality, and inter-city inequality to the overall inequality were then parsed as follows:

$$T = TWG + TBG = \sum_k \frac{p_k}{p} \left[\sum_{i \in S_k} \frac{p_i}{p_k} \log \left(\frac{p_i}{p_k} * \frac{c_k}{c_i} \right) \right] + \sum_k \frac{p_k}{p} \log \left(\frac{P_k}{P} * \frac{C}{C_k} \right) \quad (12)$$

where TWG is within-group variation, TBG is between-group variation, and p_i , p_k , and p are the individual i , group K , and overall population, respectively.

2.3.4. Driving Factors of PM_{2.5} Inequality in the Central Plains Urban Agglomeration

An interpretable machine learning approach, the Random Forest algorithm with SHapley Additive exPlanations (RF-SHAP), was used to identify and evaluate the key drivers of PM_{2.5} exposure inequality. This approach integrated various factors, including natural elements such as the NDVI and elevation; climate variables such as precipitation and temperature; emission-related data such as ground-level PM_{2.5}, EC, and O₃; and socio-economic indicators, including GDP, NTL, and POP. To ensure that the results are reliable and not affected by data dimensions, we normalized all data before applying the model to guarantee the reliability of the results.

To implement the RF-SHAP model, the RF algorithm was applied, leveraging bootstrap sampling to extract influential drivers from datasets representing overall, rural, and urban inequality. The data was divided into training and test sets with a ratio of 7:3. For each subset, decision trees were constructed, and model parameters were fine-tuned through iterative experiments. The decision tree's maximum depth was capped at 3, the step size was set to 1, the thread count was limited to 2, and the number of iterations was fixed at 100. These optimized decision trees were then aggregated to derive comprehensive results [54,55].

Shapley additive explanations is a post hoc interpretability method grounded in game theory. It quantifies the contribution of each input variable to a machine-learning model [56]. The SHAP values indicate the significance of features and clarify their directional contribution. A positive SHAP value suggests that increasing the feature enhances the predicted outcome, whereas a negative value implies the opposite [55]. Shapley additive explanations provides both global insights into model behavior and localized explanations for individual predictions. Its versatility makes it applicable across industrial, environmental, social, and economic domains [57–60]. The SHAP modelling equation is as follows:

$$f(x) = \varphi_0(f, x) + \sum_{i=1}^p \varphi_i(f, x) \quad (13)$$

$$\varphi_i(f, x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(p - |S| - 1)!}{p!} (f_x(S \cup \{i\}) - f_x(S)) \quad (14)$$

where S is a subset of the features used; p is the number of influencing factors in the model; and $f_x(S)$ represents the predicted value for subset S . The impact of the i -th factor on the prediction result of the model, given the input x , is denoted as $\varphi_i(f, x)$.

2.3.5. Spatial Concentration Effects of PM_{2.5} in the Central Plains Urban Agglomeration

A hotspot analysis was employed in this study to identify significant spatial clusters of PM_{2.5} inequalities for 2000, 2005, 2010, 2015, and 2020. It was also used to examine the PM_{2.5} exposure inequalities across the entire urban agglomeration and the inequalities between

urban and rural areas (Formula 15). The analysis utilized the Getis-Ord G_i^* statistic, which calculates a probability value (p -value) for each spatial cell by assessing the sum of values in the cell and its neighbors against the expected values under a random distribution assumption. Areas with a p -value of 0.1 or lower were categorized as hotspots or coldspots and further classified as 90%, 95%, or 99% significant [61]. This approach effectively captured spatial clustering and variability in $PM_{2.5}$ concentrations and exposure inequalities:

$$G_i^* = \frac{\sum_{j=1}^n x_j w_{i,j} - \bar{X} \sum_{j=1}^n w_{i,j}}{S * \sqrt{\left[n \sum_{j=1}^n w_{i,j}^2 - \left(\sum_{j=1}^n w_{i,j} \right)^2 \right]}} \quad (15)$$

where G_i^* is the Z-score. High Z-scores indicate hotspots, suggesting evaluated $PM_{2.5}$ concentrations, while low Z-scores indicate coldspots, indicating lower $PM_{2.5}$ concentrations. $w_{i,j}$ represents the spatial weight between subnational regions i and j . $\bar{X}$ denotes the mean $PM_{2.5}$ concentration across the region, n is the total number of subnational regions in the CPUA, and S is the standard deviation of $PM_{2.5}$ for all subnational level regions.

3. Results and Analysis

3.1. Spatio-Temporal Trends and Forecasts of $PM_{2.5}$

Figure 2 shows that from 2000 to 2023, the regions with highly significant reductions (HSRs) in $PM_{2.5}$ were mainly concentrated in Zhengzhou, the core city of the CPUA, while the significant reductions (SR) were primarily distributed along the central development corridor and southwestern regions, forming a clear clustering pattern (Figure 2a). It was projected that 99.66% of the CPUA will maintain a continuously declining trend of $PM_{2.5}$ concentrations, 0.33% will remain stable, and only 0.01% will display an upward trend in future 20 years (Figure 2b). The overall trend was dominated by significant reductions (SRs) and slightly significant reductions (SSRs), covering 0.118 million km^2 (41.02% of the total area) and 0.162 million km^2 (56.42%), respectively. In contrast, areas with no slight reduction (NSR) and no change (NC) accounted for 0.006 million km^2 (2.22%) and 0.001 million km^2 (0.32%), respectively (Figure 2c). The average $PM_{2.5}$ concentration in the CPUA exhibited a distinct inverted U-shaped trajectory, increasing from 61.50 $\mu g/m^3$ in 2020 to a peak of 88.82 $\mu g/m^3$ in 2013, before declining to 42.69 $\mu g/m^3$ in 2023 ($R^2 = 0.883$, $p < 0.001$ ***, time series regression) (Figure 2d). At the prefecture-level city scale, only Huaibei was dominated by an NSR, while all other cities were dominated by SSRs and SRs, indicating that most cities in the CPUA were experiencing a continuous downward trend in $PM_{2.5}$ concentrations (Figure 2e).

3.2. Regional Inequalities in the $PM_{2.5}$ Concentration

Figure 3 shows that from 2000 to 2020, the overall Theil index of the CPUA decreased from 0.168 to 0.142, indicating a reduction in $PM_{2.5}$ exposure inequality across the region. A significant decline was observed from 2011 to 2012, with the index dropping from 0.174 to 0.141. Within-group variation remained stable at approximately 0.043, while TBG decreased from 0.121 to 0.098, with a sharp drop from 0.130 to 0.099 during 2011–2012 (Figures 3a and S3 and Tables S1–S3).

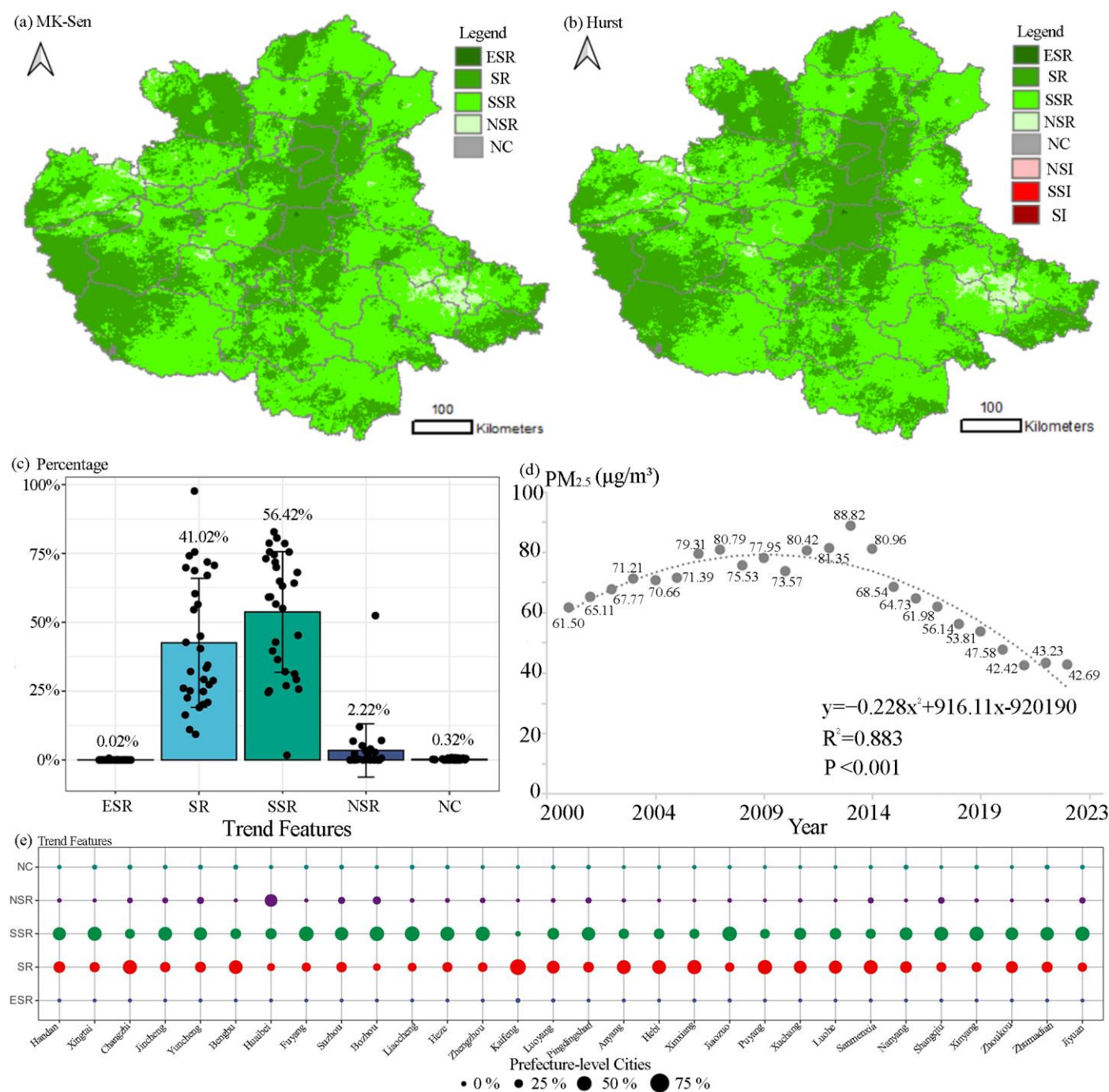


Figure 2. Trends in PM_{2.5} intensity and spatiotemporal evolution across the Central Plains Urban Agglomeration (CPUA). (a). Sen-MK test results showing the PM_{2.5} concentration trends. (b). Future PM_{2.5} concentration trends derived from the Hurst index. (c). Percentage distribution of the five PM_{2.5} trend categories across cities. (d). Interannual variation in the average PM_{2.5} concentration. (e). Spatial distribution of PM_{2.5} concentration trends across 30 prefecture-level cities.

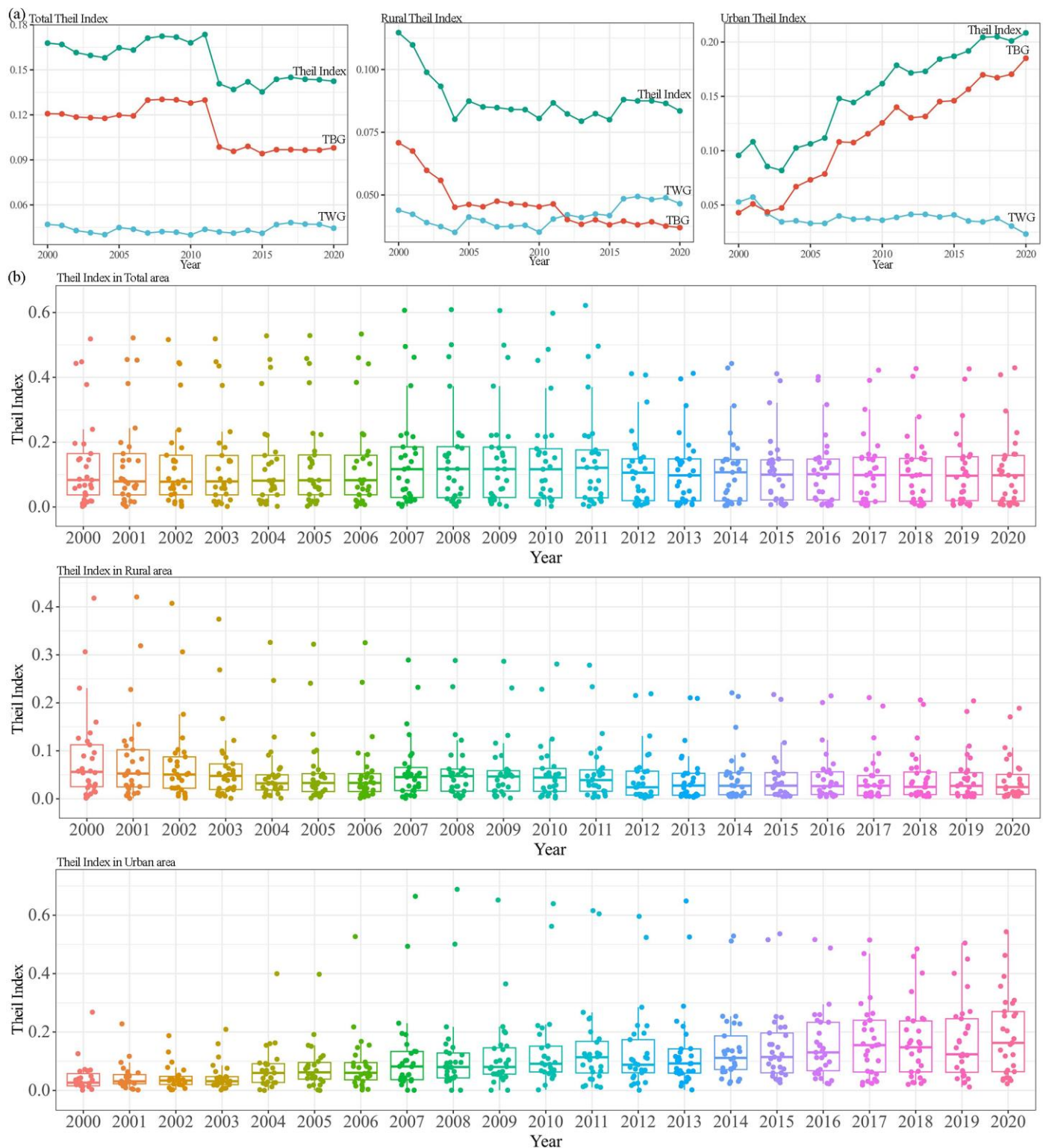


Figure 3. Spatial and temporal characteristics of PM_{2.5} inequality. (a). Theil indices for the whole region, rural, and urban areas. (b). Temporal distributions of the Theil indices from 2000 to 2020.

To further investigate air pollution exposure inequalities between urban and rural areas, the Theil indices were determined for these regions. In rural areas, the total Theil index decreased from 0.115 to 0.084. Between-group variation decreased significantly from 0.071 to 0.037, whereas TWG increased slightly from 0.044 to 0.047. Consequently, TWG's contribution to overall inequality increased from 38% to 56%. Notably, in 2012, TWG surpassed TBG as the dominant factor causing inequality. In general, these phenomena

indicated a growing disparity in rural PM_{2.5} exposure between cities. In urban areas, the total Theil index steadily increased from 0.096 to 0.208. A decomposition analysis of Theil index revealed a decline in TWG from 0.053 to 0.023, whereas there was an increase in TBG from 0.043 to 0.185. The contribution of TBG to the overall Theil index increased substantially, from 45% to 89%, indicating that inter-city disparity in PM_{2.5} exposure within urban areas continued to expand. In summary, the rural and overall inequalities of PM_{2.5} exposure have decreased with the growing disparity between cities becoming the primary driver of regional inequality in the CPUTA (Figure 3b and Figure S5).

3.3. The Differences in the Driving Factors of Urban–Rural PM_{2.5} Exposure Inequality

This study explored the inequalities in the drivers of PM_{2.5} exposure inequality across the CPUTA as a whole and between urban and rural areas in 2000, 2005, 2010, 2015, and 2020. Figure 4a,b and Table S8 show that according to the average SHAP values for each of the five years, the factors with the greatest explanatory power for the inequality in PM_{2.5} exposure, from strongest to weakest, were as follows: elevation (0.593), electricity consumption (0.433), ground-level PM_{2.5} (0.318), population density (0.258), GDP (0.256), O₃ concentration (0.204), NDVI (0.155), temperature (0.116), NTL (0.109), and precipitation (0.062). These results indicated that natural factors (i.e., elevation) had the strongest explanatory power for the inequality of overall PM_{2.5} exposure in the CPUTA, while the impact of socio-economic covariates could not be ignored. In particular, electricity consumption should be given more attention (Figure 4a,b).

In rural areas of the CPUTA, since most variables show significant differences in performance between the training set and the test set, most drivers had no explanatory power, with only elevation having notable SHAP values of 0.184, 0.392, 0.173, and 0.178 in 2000, 2005, 2015, and 2020, respectively, and temperature having a SHAP value of 0.221 in 2010. This indicated that consistent with the CPUTA as a whole, natural factors (i.e., elevation) were the most significant explanatory factor for inequality in PM_{2.5} exposure in rural areas of the CPUTA (Figure S1 and Table S8).

Figure 4c,d shows that in urban areas, population density consistently exhibited the strongest explanatory power throughout the study period, with mean values of 0.553, 0.555, 0.553, 0.520, and 0.482 in 2000, 2005, 2010, 2015, and 2020, respectively. Elevation was ranked as the second most influential factor, with average values of 0.324, 0.315, 0.306, 0.308, and 0.295 in 2000, 2005, 2010, 2015, and 2020, respectively. From the average SHAP values for the five years, the factors with the greatest explanatory power for inequality in PM_{2.5} exposure, from strongest to weakest, were the following: population density (0.534), elevation (0.310), O₃ concentration (0.218), GDP (0.211), electricity consumption (0.195), precipitation (0.192), NTL (0.081), ground-level PM_{2.5} (0.059), temperature (0.057), and NDVI (0.004). Unlike the overall and rural areas of CPUTA, the inequality of PM_{2.5} exposure in the urban areas of the CPUTA was mainly influenced by factors such as population density, GDP, and electricity consumption. However, elevation still had a relatively strong influence. The O₃ concentration should be managed in synergy with PM_{2.5} because of its strong explanatory power (Figure 4c,d).



Figure 4. The SHapley Additive exPlanations (SHAP) values in 2000, 2005, 2010, 2015, and 2020. (a). Average SHAP values for overall $PM_{2.5}$ exposure inequality within the CUPA. (b). A bee swarm plot of the individual SHAP values for overall $PM_{2.5}$ exposure inequality within the CUPA. The y-axis stands for the different driving factors. (c). Average SHAP values for $PM_{2.5}$ exposure inequality in urban areas within the CUPA. (d). A bee swarm plot of the individual SHAP values for $PM_{2.5}$ exposure inequality in urban areas within the CUPA. The y-axis stands for the different driving factors. (Note vegetation cover (NDVI), elevation (ELE), precipitation (PRE), temperature (TEM), $PM_{2.5}$ (PM), electricity consumption (EC), ozone concentration (O_3), nighttime lighting (NTLI), population density (POP), and gross domestic production (GDP)).

3.4. Identifying Hotspots of $PM_{2.5}$ Exposure Inequality

Figure 5 showed that from 2000 to 2020, the proportion of $PM_{2.5}$ -exposed coldspot areas decreased from 34.14% to 30.89%, while the proportion of hotspot areas increased from 31.08% to 36.60%. The coldspot areas were primarily located in the northwestern and southwestern regions of the CUPA. The hotspot areas were predominantly concentrated in the north-central part, with the remaining areas showing no significant changes. The

hotspot areas in northeastern Zhoukou, Shangqiu, and Heze further expanded in size, and the hotspot areas in the north-central part of the CPUA spread further to the southeast. In Yuncheng in the northwestern part of the CPUA, there was a shrinkage of the coldspot area and the emergence of a hotspot area in 2020. At the same time, the coldspot areas in Fuyang, Xinyang, Bozhou, and Bengbu in the southwestern part of the CPUA declined further (Figures 5a and S2). During the same periods, the overall hotspot areas based on the Theil index were largely concentrated in Changzhi, Jincheng, Luoyang, and Jiaozuo, reflecting an unequal spatial distribution of $PM_{2.5}$ pollution exposure (Figure 5b). In rural areas, the Theil index hotspots were mainly observed in Changzhi, Jincheng, and Jiaozuo. Bozhou transitioned into a coldspot in 2015, and by 2020, Bozhou, Huaibei, and Shangqiu formed a coldspot cluster, signifying a spatial aggregation of regions with high equality in $PM_{2.5}$ exposure in rural areas (Figure 5c). The Theil index values for urban areas did not have obvious spatial clustering characteristics, indicating that the characteristics of $PM_{2.5}$ exposure inequality in the urban areas of the CPUA prefectural administrative divisions were more dispersed, which makes them unsuitable for collaborative management strategies (Figure 5d).

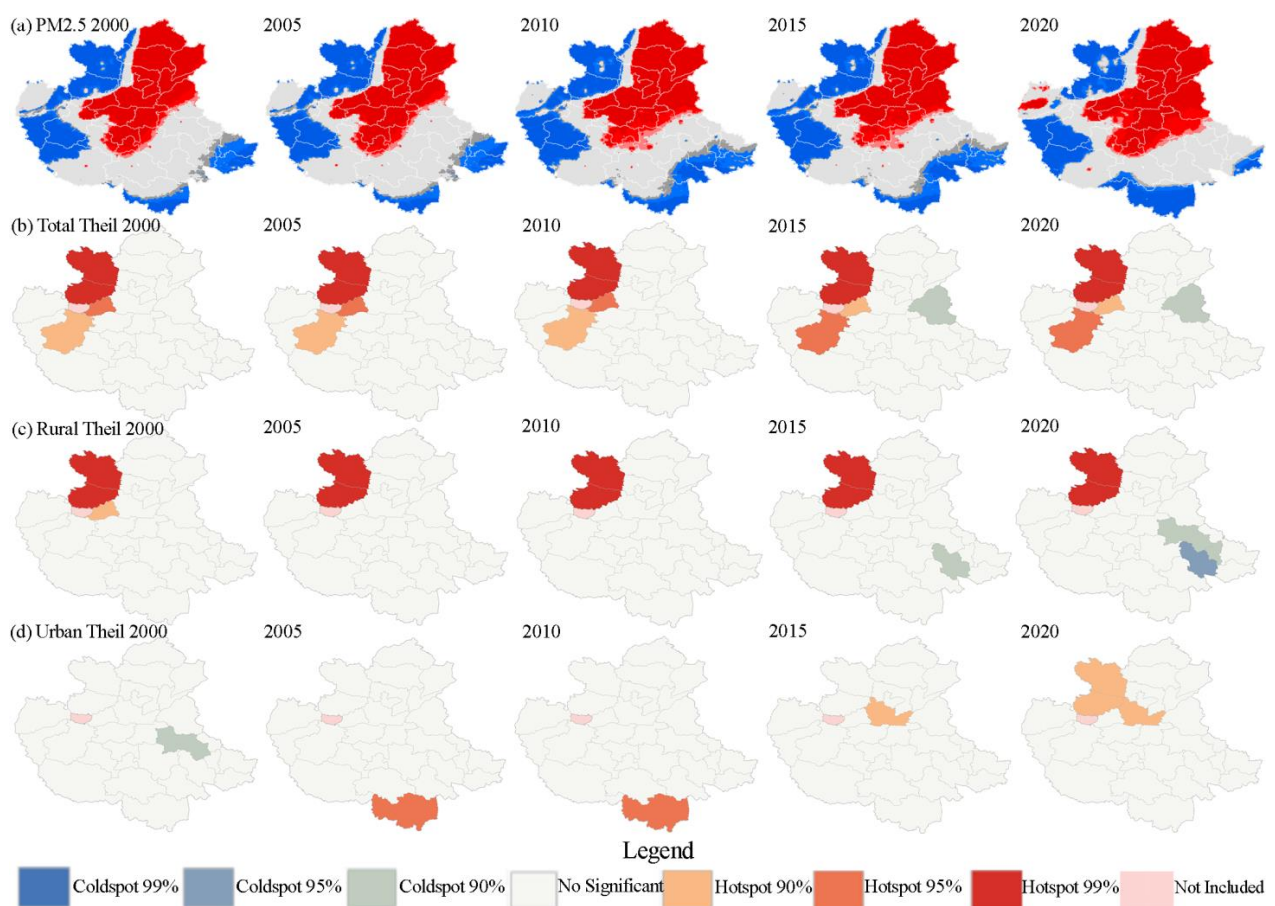


Figure 5. Spatial hotspot–coldspot distribution of $PM_{2.5}$ concentration and inequality in the Central Plains Urban Agglomeration (CPUA) every five years from 2000 to 2020. (a). Spatial hotspot–coldspot distribution of $PM_{2.5}$ concentration at the grid scale (1×1 km), where light orange, orange-red, and red represent hotspots with confidence levels of 90%, 95%, and 99%, respectively. Grey, light blue, and dark blue represent coldspots with confidence levels of 90%, 95%, and 99%, respectively; white represents areas without significant change. (b). Spatial hotspot–coldspot distribution of the overall $PM_{2.5}$ Theil index at the prefecture-level city scale. (c). Spatial hotspot–coldspot distribution of the $PM_{2.5}$ Theil index in rural areas at the prefecture-level city scale. (d). Spatial hotspot–coldspot distribution of the $PM_{2.5}$ Theil index in urban areas at the prefecture-level city scale.

4. Discussion

4.1. Spatial and Temporal Change of $PM_{2.5}$ Exposure in CPUA

During the study period, the $PM_{2.5}$ concentrations in the CPUA initially exhibited an upward trend, followed by a marked decline after reaching an inflection point in 2013. This sustained decrease from 2013 to 2020 was closely associated with the implementation of the Air Pollution Prevention and Control Action Plan introduced by the Chinese government in 2013. The plan effectively curtailed $PM_{2.5}$ levels by shutting down heavily polluting enterprises, enforcing strict pollution controls, and promoting environmentally friendly pollutant treatments [34,46]. Cities such as Anyang, Hebi, Xinxiang, Zhengzhou, Xuchang, Zhumadian, and Xinyang, situated within the central development corridor of the CPUA, were historically characterized by early industrialization and the dominance of heavy industry. Consequently, these regions experienced a significant reduction in $PM_{2.5}$ levels post-2013 due to stringent policy measures. Notably, Zhengzhou, the central city of the CPUA, experienced a highly significant decline in $PM_{2.5}$ concentrations. Zhengzhou City implemented rigorous pollution control policies. However, it also faces persistent challenges from the urban heat island effect, which facilitates pollutant transport from surrounding areas to urban regions. Some studies have reported that $PM_{2.5}$ concentrations are strongly correlated with the magnitude of the urban heat island effect. Furthermore, $PM_{2.5}$ in the CPUA exhibits obvious spatial spillover and agglomeration effects, presenting significant challenges to the collaborative management of pollution between rural and urban areas, and within urban agglomerations [62,63]. Nevertheless, China's air pollution control policy measures have resulted in continuous improvements in air quality within the CPUA [14].

4.2. Rural–Urban Differences in $PM_{2.5}$ Exposure

The study revealed a declining trend in the overall inequality of $PM_{2.5}$ pollution exposure across the CPUA, with a sharp reduction observed from 2011 to 2012. This significant improvement was largely attributed to the Chinese government's 2011 Joint Air Pollutant Prevention and Control Policy, which promoted a coordinated approach to control regional air pollution. As $PM_{2.5}$ exposure inequalities among cities narrowed, regional inequality in pollution exposure diminished, facilitating collaborative air pollutant management within urban agglomeration.

In rural areas, $PM_{2.5}$ pollution exposure has steadily decreased from 2000 to 2004, stabilizing after 2005. This trend was aligned with the urbanization process in the CPUA, marked the fastest pace of urbanization, driven by the “New Rural Construction” policy and rural-to-urban migration [15,16]. These developments reduced variability in $PM_{2.5}$ exposure in rural areas, improving public health and promoting equitable regional development [64].

Conversely, $PM_{2.5}$ pollution exposure inequality in urban areas has steadily increased, a trend closely associated with the rapid pace of urban land development. This process has resulted in imbalanced regional development, leading to concentrated populations in heavily polluted urban and industrial zones [15]. This has increased urban residents' exposure to air pollution. The failure of rapid urbanization to safeguard public health, ensure equitable development, and bridge economic inequalities among cities has perpetuated a vicious cycle [7]. Consequently, the inequalities in $PM_{2.5}$ pollution exposure among urban areas have intensified, further exacerbating regional inequalities [65].

4.3. Differences in the Drivers of Inequality in $PM_{2.5}$ Exposure Between Urban and Rural Areas

The RF-SHAP results revealed that elevation had the greatest explanatory power for both overall and rural $PM_{2.5}$ exposure in the CPUA. It means that elevation was identified

as the most influential factor contributing to the inequality of PM_{2.5} exposure in the CPUA. Elevation influences local wind patterns and terrain-induced airflow dynamics [66]. Low-lying areas are more prone to becoming pollutant convergence zones due to topographical constraints, which, when coupled with the urban heat island effect, can exacerbate the spatial disparity in PM_{2.5} exposure. Additionally, the transport of air pollutants is modulated by terrain-induced elevation differences. High-elevation areas are more likely to receive long-range transported pollutants from the upper atmosphere, while low-elevation regions primarily accumulate locally emitted pollutants [67,68].

Population density was identified as the strongest explanatory factor for regional inequality within the CPUA urban area, which suggests that the population density of residential areas may result in PM_{2.5} exposure inequalities [14]. In China's urban agglomerations, vulnerable communities, such as impoverished neighborhoods and urban villages that often lack essential infrastructure (e.g., education and healthcare), typically have low population densities [22]. Conversely, areas with higher population densities tend to have more comprehensive infrastructure, making population density a crucial determinant in assessing inequalities in urban PM_{2.5} exposure [21,23]. Electricity consumption and GDP also significantly affected urban PM_{2.5} exposure inequalities. GDP exerted a dual impact during the initial stages of economic growth because it drove rapid regional development at the expense of environmental equality. Conversely, in the advanced stages of economic development, it fostered growth by encouraging the adoption of cleaner and more sustainable production practices [69]. It therefore alleviated PM_{2.5} exposure inequalities within the CPUA. Notably, the O₃ concentration also had a substantial explanatory power for PM_{2.5} exposure inequalities, indicating an urgent need for integrated management strategies for both O₃ and PM_{2.5} within the CPUA to effectively address environmental pollution inequalities [70].

4.4. Limitations of the Study and Future Outlook

The main limitation of this study was the spatial and temporal resolution of the data sources. The analysis relied on annual air pollution exposure data, and it was therefore difficult to incorporate higher temporal resolution data (e.g., daily or monthly PM_{2.5} levels) [34]. Additionally, the study's temporal scope (2000–2023) and urban agglomeration scale necessitated the use of 1 × 1 km resolution raster data, which may have introduced underestimation or overestimation errors due to geospatial data uncertainties and potential spatial inaccuracies.

The second limitation was the exclusive use of the Theil index to calculate regional air exposure inequality at the municipal scale. While this approach is valid, previous studies employed alternative measures such as the Gini index, Hoffman index, and Atkinson index to assess regional development inequality [7,14]. The absence of these additional indices may have led to inaccuracies in the calculation of inequality. Furthermore, the study did not account for atmospheric dynamics and spatial diffusion effects, potentially overlooking cross-boundary interactions, such as the diffusion of air masses between Zhengzhou and Kaifeng in the CPUA [64], which could have affected air exposure inequality calculations at the prefecture-level scale.

Last, to maximize the capacity for capturing long-term trends in PM_{2.5} exposure inequality, we employed annual PM_{2.5} concentration data at a spatial resolution of 1 km. This choice was guided by the need to balance temporal coverage and spatial detail over an extended time series. Although higher temporal resolution data (e.g., monthly or daily) could potentially offer more detailed insights, previous studies have indicated that temporal variations in inequality at these finer scales are relatively minor.

To address these limitations, we propose several recommendations for future research. First, higher temporal resolution data, such as daily PM_{2.5} levels, could be used to focus on the number of polluted days rather than annual averages, mitigating the influence of extreme exposure days. Second, future studies should consider quantifying air inequality at the raster scale and analyzing regional air exposure inequalities across broader spatial scales. This approach would overcome the constraints of administrative boundaries, enabling a more precise assessment of inequalities within urban agglomerations. Finally, combining multiple inequality indices to measure air pollution exposure would help identify the most appropriate metric for urban agglomeration scales, fostering more effective collaborative air pollution management strategies.

5. Conclusions

This study used multi-source geospatial data to meticulously examine the trends in PM_{2.5} concentrations across the CPUA from 2000 to 2023 and then projected future patterns. This study answered three scientific questions:

(1) How have the spatial and temporal patterns of PM_{2.5} pollution evolved within representative urban agglomerations, and what insights can be derived from forecasting future trends and identifying pollution hotspots? A consistent downward trend in PM_{2.5} concentrations across the region was identified, with future trends expected to align with this trajectory. From 2000 to 2020, almost all of the CPUA exhibited a decreasing trend of PM_{2.5}, which is in line with the Chinese government's national strategy of "reducing the concentration of air pollutants". Additionally, cold spots of PM_{2.5} exposure were concentrated in the northwestern and southwestern areas of the region, whereas hotspots were primarily located in the north-central areas.

(2) What are the dynamics of urban–rural PM_{2.5} exposure inequalities within urban agglomerations, and how have they changed over time? From 2000 to 2020, PM_{2.5} exposure inequalities decreased in both overall and rural areas but escalated in urban settings. Notably, Changzhi, Jincheng, Luoyang, and Jiaozuo were identified as the most significant hotspots of PM_{2.5} exposure inequality, underscoring the urgent need for integrated management strategies in these regions.

(3) How do natural, climate, and socio-economic drivers collectively influence urban–rural exposure inequalities, and what strategies can be adopted for effective mitigation? The overall PM_{2.5} exposure inequality and urban–rural inequalities were systematically analyzed, and the underlying drivers of spatial hotspots of exposure inequality were determined. Elevation emerged as a pivotal factor influencing exposure inequality in overall and rural areas, while population density was the main dominant factor in densely populated urban regions.

The main achievements of this study included accurately mapping the spatiotemporal dynamics of air pollution exposure within the CPUA; revealing regional inequalities in exposure inequality (at overall, urban, and rural scales); highlighting the critical need for intensified air pollution management in urban areas; and emphasizing the need to reduce inter-regional inequalities in exposure to achieve SDGs (SDG 3; SDG 10, and SDG 11). Furthermore, it quantified the driving forces of exposure inequality, providing localized, valuable evidence and actionable policy recommendations for the coordinated management of PM_{2.5} exposure inequalities. Ultimately, the findings of this study will advance sustainable development and foster social equity within the region.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/rs17172982/s1>. Supplementary material to this article included the Supplementary Discussion, the Figures S1–S6, and the Tables S1–S8 [71–78].

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Data Availability Statement: Research Data can be found at Supplementary Material Table S1–S8, origin data can be found at Table 1, other data are available upon request from the corresponding author.

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